Case Study: Maximizing e-Bike Motor Performance through Stator Flux- Barrier Innovation

Client: E-Bike OEMs & E-Mobility Engineering Firms

Prepared by: FEAAM GmbH

Subject: High Power Density e-Bike Traction Motors via Stator Flux-Barrier Technology

Executive Summary

In the rapidly growing e-mobility sector, the demand for high-torque, compact Propulsion systems have hit a technical ceiling with traditional motor designs. FEAAM GmbH has pioneered a breakthrough by integrating stator flux-barriers—strategically engineered slits in the motor core—to overcome the inherent limitations of standard concentrated windings. This innovation achieves a 16% boost in the working magnetic harmonic and a 73% reduction in harmful subharmonics without increasing motor size or material costs. By bridging the gap between advanced electromagnetic research and mass manufacturing, FEAAM enables OEMs to deliver superior power density, enhanced cooling, and a premium, vibration-free riding experience.

The Challenge: The FSCW Performance Bottleneck

Fractional slot concentrated windings (FSCW) are the preferred choice for e- bikes due to their low manufacturing costs, high slot fill factors, and compact end windings. However, they present a significant "purity" problem:

- Magnetic Interference: These windings generate a non-sinusoidal magnetic field (MMF) filled with parasitic spatial harmonics.
- Parasitic Losses: Unmanaged harmonics (especially the 1st and 5th) lead to localized core saturation and excessive heat in the permanent magnets.
- Operational Limits: These issues cause Noise, Vibration, and Harshness (NVH), which forces engineers to trade off high performance for thermal stability and ride quality.

The FEAAM Innovation: Stator Flux-Barrier (FB) Technology

Instead of complex and expensive winding modifications, FEAAM utilizes structural optimization of the stator core. By applying magnetic flux-barriers directly in the stator teeth, we effectively "filter" the magnetic field at its source.

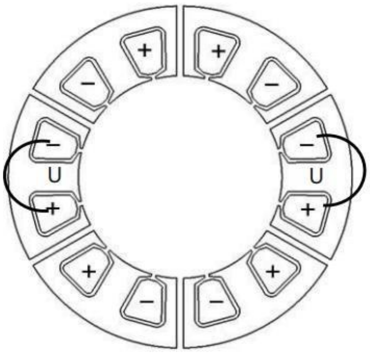

Fig. 1. Flux-barrier stator

Strategic Advantages:

- Harmonic Filtering: Our flux-barriers act as highway dividers, steering the magnetic "traffic" into productive lanes while blocking the "noise" that causes energy loss.
- Efficiency Foundation: This allows the use of Single-Layer (SL) windings, which have a superior winding factor of 0.966 for the working harmonic compared to only 0.933 for standard double-layer designs.
- Secondary Benefits: The barrier regions can be utilized for cooling fins or optimized for even higher slot fill factors, ensuring the motor runs cooler under high loads.

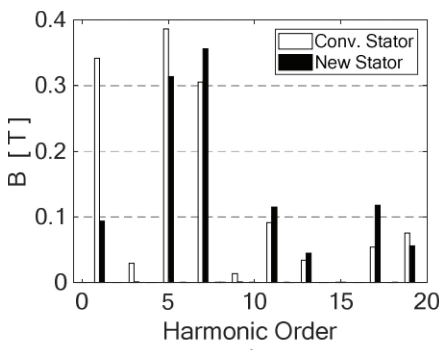

Fig. 2. Comparison of air-gap flux density harmonics due to stator winding.

Practical Application: 20-Pole e-Bike Traction Motor

FEAAM validated this technology through an optimized 12-teeth/10-pole pairs SL configuration designed for high-end traction performance.

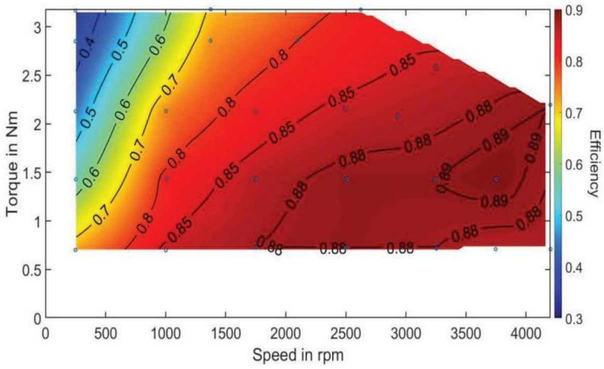

Fig. 3. Efficiency map for the flux-barrier e-bike motor.

Description	Specification	Benefit
Peak Efficiency	90%	Maximum battery range
Peak Torque	3.25 Nm	Superior climbing power
Torque Ripple	4%	Smooth, vibration-free ride
Thermal Constant	30 minutes	Sustained high-load durability

The Bottom Line for Business

FEAAM's proprietary stator architecture allows manufacturers to bypass traditional design trade-offs.

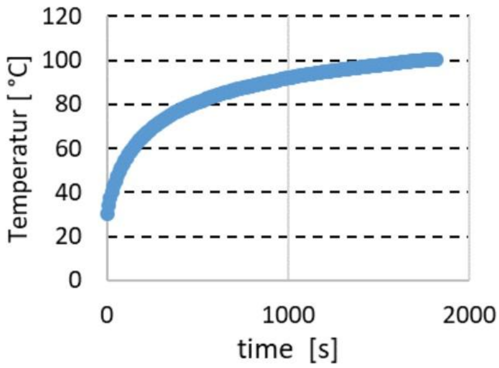

Fig. 4. Winding temperature with the time at 2,33Nm and 2500rpm.

By partnering with us, OEMs can:

- **Reduce Production Costs:** Retain the cost-saving benefits of simple single-layer concentrated windings while achieving performance that typically requires complex, expensive alternatives.
- **Extend Component Life:** A **73% reduction in the first subharmonic** drastically lowers rotor losses and prevents premature component failure.
- **Differentiate Through Quality:** Deliver motors that are not only more powerful but significantly quieter and smoother, meeting the demands of the premium e-mobility market.

FEAAM can assist your team by integrating these strategically engineered stator flux-barriers into your concentrated winding designs, reducing harmful subharmonics by 73% to deliver a highly efficient, vibration-free riding experience without increasing your manufacturing or material costs.